

DAY 6

Model Audit and Risk Management

FMVA Certification Prep: Financial Modeling and Valuation Bootcamp

Lesson Forge Pro (owner)

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This chapter builds working competence in Model Audit and Risk Management: the concepts, the standard procedure, and the judgment to apply it under exam conditions.

Concepts and vocabulary

Core Definitions

Model audit and risk management sits at the intersection of quantitative finance, governance, and quality assurance. Before examining techniques, you need a precise shared vocabulary—ambiguous terminology is itself a source of model risk.

Model — a quantitative representation of a financial relationship, used to produce outputs (prices, risk metrics, forecasts) from inputs and assumptions. Examples range from a discounted cash flow (DCF) spreadsheet to a stochastic volatility option pricer.

Model risk — the potential for adverse consequences arising from decisions based on incorrect or misused models. Regulators (SR 11-7 / SS3/18) decompose this into two sources: fundamental model error (wrong theory or mathematics) and implementation error (correct theory, faulty code or data).

Model audit — an independent, structured review that assesses whether a model is conceptually sound, correctly implemented, and used appropriately. It produces a formal opinion and a findings register.

Model validation — often used interchangeably with model audit in practice, though strictly validation is the ongoing, owner-side process; audit implies independence from the development team.

Model inventory — the firm-wide register of all models in production, recording owner, use case, validation status, and materiality tier.

Materiality tier — a risk-based classification (commonly Tier 1 / High, Tier 2 / Medium, Tier 3 / Low) that determines review frequency and depth.

Spreadsheet risk — a subset of model risk specific to workbook-based models: broken links, hardcoded values masquerading as formulas, version-control failures, and unprotected inputs.

Where It Fits in Practice

Model audit is a second-line-of-defence activity. The model developer (first line) builds and documents; the independent validation or audit team (second line) challenges; internal audit (third line) checks that the governance framework itself is functioning. In quantitative finance, audited models include option pricers (accommodating volatility surfaces and multi-curve frameworks), XVA engines, credit risk models (PD, LGD, EAD for regulatory capital), and portfolio optimisation tools.

Good practice—drawn from professional guidelines—requires that models separate input, calculation, and output layers to enhance traceability. Auditors test each layer independently.

Notation

| Symbol | Meaning |

|-----|-----|

| M | Model under review |

| I | Input set (data, parameters, assumptions) |

| $f(\cdot)$ | Functional mapping (the model logic) |

| O | Output: $O = f(I)$ |

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| MR | Model risk score (materiality × likelihood of error) |

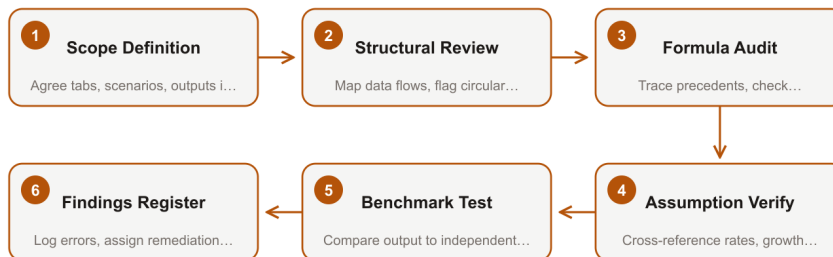
The fundamental audit identity is simply:

$$O = f(I) \pm ;P$$

Every audit procedure is ultimately trying to bound or explain ;Rà

Worked Example — Terminology Applied to a Mini Scenario

Model Audit Standard Procedure



Every audit follows this repeatable sequence regardless of model type

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Scenario: A corporate treasury team uses an Excel-based DCF model to value a five-year infrastructure project. The risk-free rate is 3.2 %, the equity risk premium is 5.5 %, and the project beta is 0.80. WACC is calculated as the blended cost of capital.

Step 1 — Identify the model (M).

M is the DCF workbook. Its stated purpose is project valuation for capital budgeting approval.

Step 2 — Enumerate the input set (I).

$I = \{\text{risk-free rate} = 3.2\%, \text{ERP} = 5.5\%, \text{FV's weight} = 40\%, \text{cost of debt} = 4.0\%, \text{tax rate} = 25\%, \text{five years of projected free cash flows}\}.$

Step 3 — Trace the functional mapping $f(\cdot)$.

The model computes cost of equity via CAPM:

$$K_e = R_f + \beta \times \text{ERP} = 3.2\% + 0.85 \times 5.5\% = 7.6\%$$

Then WACC:

$$\text{WACC} = K_e \times (1 - w_d) + K_d \times (1 - t) \times w_d = 7.6\% \times 0.60 + 4.0\% \times 0.75 \times 0.40 = 4.56\% + 1.20\% = 5.76\%$$

Step 4 — Assign materiality tier.

The project NPV is £42 million against a firm threshold of £10 million for Tier 1. The model is classified Tier 1 — High, requiring full independent audit annually.

Step 5 — Measure model error; R^2

The auditor re-runs the WACC using a Bloomberg consensus beta of 0.85 (the developer used a five-year monthly regression). The revised WACC is $5.76\% + (0.05 \times 5.5\% \times 0.60) = 5.93\%$. The NPV falls by £1.4 million. $R^2 = 0.36$ finding logged in the findings register.

Step 6 — Record in model inventory.

The model inventory entry is updated: owner = Treasury, tier = 1, last audit = current date, open findings = 1 (beta source), status = Conditionally Approved — remediation required within 30 days.

Common mistake: Auditors sometimes test only the formula logic and overlook input sourcing. In this example, the formula was correct—the error lived entirely in the data feed. Always trace inputs back to their primary source, not just the cell reference within the workbook.

The standard procedure

A model audit applies a structured, repeatable procedure to verify that a financial model is technically correct, logically sound, and fit for its intended purpose. Whether the model prices an interest rate swap, projects free cash flow for a DCF valuation, or calculates regulatory capital under a credit risk framework, the same disciplined sequence of inputs, steps, outputs, and checks applies every time.

Inputs

Before any testing begins, the auditor assembles the following:

- The model file itself — typically an Excel workbook, including all sheets, named ranges, and any linked external files.
- Model documentation — the specification, assumption log, and version history.
- Source data — market data feeds, accounting statements, or parameter tables used to populate the model.
- Prior audit reports or validation notes, if available.

- A benchmark or independent calculation — even a simple back-of-envelope figure against which outputs can be compared.

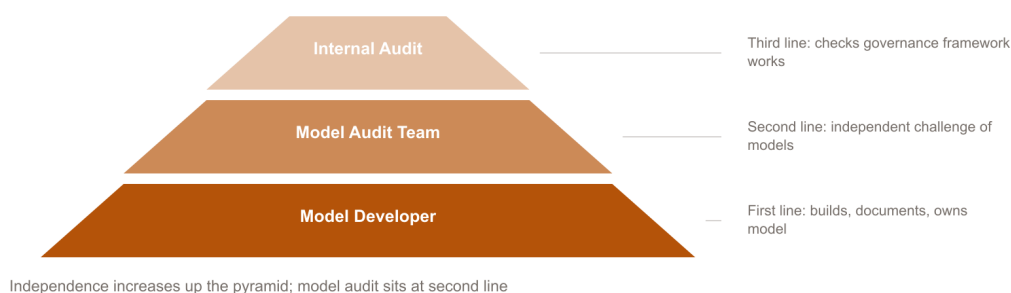
Separating input sheets from calculation and output sheets (a best-practice standard widely endorsed in professional guidelines) makes this intake step far easier and reduces the risk of overlooking a hard-coded assumption buried inside a formula.

Step-by-Step Procedure

- Scope definition. Agree with the model owner on what the audit covers: which tabs, which scenarios, which outputs are in scope.
- Structural review. Map the model's architecture — data flows from inputs through calculations to outputs. Flag circular references, merged cells, and hidden rows.
- Formula audit. Trace precedents and dependents for every key formula. Check that cell references are consistent and that no values are hard-coded where a parameter cell should be used.
- Assumption verification. Cross-reference every assumption (discount rate, growth rate, default probability, etc.) against the documented source.
- Independent recalculation. Rebuild critical outputs from scratch using the same inputs. Compare results to the model's figures.
- Stress and scenario testing. Apply extreme but plausible inputs (e.g., zero growth, doubled discount rate) and confirm the model responds sensibly — no errors, no implausible sign changes.
- Output review. Check that reported figures are correctly linked from calculation sheets and that rounding, units, and sign conventions are consistent throughout.
- Documentation of findings. Record every issue with its location, severity (critical / significant / minor), and recommended remediation.

Worked Numerical Example

Three Lines of Defence



Independence increases up the pyramid; model audit sits at second line

Scenario: You are auditing a simple DCF model that values a three-year project. The model claims a Net Present Value (NPV) of \$1,842.

Inputs provided:

| Year | Free Cash Flow (FCF) |

|-----|-----|

| 1 | \$800 |

| 2 | \$900 |

| 3 | \$1,000 |

Discount rate (WACC): 12%. Initial investment: \$1,500.

Step 3 — Independent recalculation:

The present value formula for each cash flow is:

$$PV = FCF \div (1 + r)^t$$

- Year 1: $800 \div (1.12)^1 = 800 \div 1.1200 = \714.29
- Year 2: $900 \div (1.12)^2 = 900 \div 1.2544 = \717.47
- Year 3: $1,000 \div (1.12)^3 = 1,000 \div 1.4049 = \711.78

Sum of PVs: $\$714.29 + \$717.47 + \$711.78 = \$2,143.54$

NPV: $\$2,143.54 - \$1,500.00 = \$643.54$

Finding: The model reports \$1,842, but the correct NPV is \$643.54 — a difference of \$1,198.46.

Tracing back through the formula audit (Step 3) reveals that the model forgot to subtract the initial investment, treating the sum of discounted cash flows as the NPV. This is a critical error.

Outputs

The audit delivers:

- A findings log listing every error, its cell location, severity, and fix.
- A corrected or annotated model file (or a formal sign-off if no issues are found).
- An audit opinion — a brief written conclusion on whether the model is fit for purpose.

Checks

Common mistake: Auditors often limit independent recalculation to a single output cell and assume the rest of the model is correct. Always recalculate each intermediate subtotal separately; errors frequently cancel each other out at the top level while remaining dangerous in scenario branches.

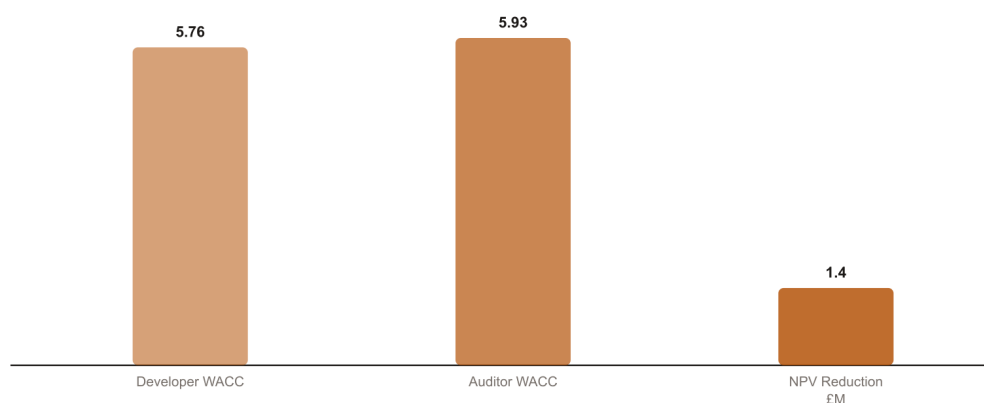
Final checks include confirming that all findings have been resolved or formally accepted, that the corrected model reproduces the auditor's benchmark figures exactly, and that the assumption log has been updated to reflect any changes made during remediation. Only after these checks pass should the model be released for decision-making use.

Applications and pitfalls

Model audit and risk management sits at the intersection of technical rigor and professional judgment. Understanding where financial models are legitimately applied—and where practitioners routinely go wrong—is essential for anyone preparing to work with or review quantitative models in practice.

Typical Use Cases

WACC Audit Error Impact



Input sourcing error (beta), not formula error, caused £1.4M NPV difference

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Financial models serve a remarkably broad range of purposes. In capital markets, they underpin option pricing (including calculation of the Greeks and accommodation of volatility surfaces through local or stochastic volatility models), interest rate derivative valuation, credit derivative pricing, and the full suite of XVA adjustments including CVA. Credit risk applications span regulatory capital calculations (EAD, PD, LGD, PFE), consumer credit scoring, IFRS 9 impairment provisioning, and structural models such as Merton and KMV.

On the corporate side, models support discounted cash flow (DCF) valuation, WACC calculations, scenario planning, budgeting, and treasury management. Portfolio optimization, real options analysis, and actuarial applications such as Dynamic Financial Analysis round out the landscape.

Despite this variety, the vast majority of these models live in Microsoft Excel spreadsheets—company-specific, analyst-built, and subject to what practitioners call spreadsheet risk. Best practice calls for separating input, calculation, and output sheets; consistent formatting; clear labeling; and thorough documentation of assumptions. These conventions exist precisely because they make models auditable and reduce the chance of silent errors propagating through complex workbooks.

Common Mistakes

Errors in financial models tend to cluster around a handful of recurring themes:

- Hard-coded values buried in formulas — a discount rate typed directly into a cell formula rather than referenced from an input sheet.
- Circular references left unresolved — particularly dangerous in iterative XVA or working capital models.
- Inconsistent time conventions — mixing annual and semi-annual compounding, or confusing 30/360 with Actual/365 day-count conventions.
- Failure to stress-test assumptions — accepting base-case outputs without checking behavior under adverse scenarios.
- Omitting sign conventions — cash outflows and inflows treated inconsistently across linked sheets.

Common mistake: Analysts frequently discount cash flows using a rate expressed in annual terms while the cash flow periods are quarterly, producing a valuation that is systematically too high. Always confirm that the discount rate period matches the cash flow period before computing present values.

Sanity Checks

Before any model output is relied upon, apply these checks:

- Order-of-magnitude test: Does the result fall within a plausible range given industry benchmarks?
- Limiting-case test: Set growth to zero and verify the model collapses to a simple perpetuity formula.
- Sign check: Confirm that higher discount rates produce lower present values, not higher.
- Sum-of-parts reconciliation: Ensure that disaggregated outputs sum to the consolidated total.
- Sensitivity table: Vary one key input (e.g., WACC $\pm 1\%$) and confirm the direction and magnitude of change are economically intuitive.

Worked Example: A Flawed DCF Corrected Step by Step

Scenario: An analyst values a project with four annual free cash flows and a terminal value.

Flawed attempt — the analyst's original inputs:

Year	FCF (£000)
1	120
2	135
3	150
4	165

Terminal value at end of Year 4 (Gordon Growth Model, $g = 2\%$):

$$TV = FCF_{\dots} / (r - g)$$

The analyst uses $r = 8\%$ (annual WACC) but discounts each cash flow by dividing by $(1 + 0.08/4)^n$, treating the rate as quarterly. This is the error.

Step 1 — Identify the mismatch. Cash flows are annual; the rate must also be applied annually. The correct discount factor for Year n is $1 / (1.08)^n$.

Step 2 — Compute $FCF_{\dots}$ and terminal value.

$$FCF_{\dots} = 165 \times 1.02 = \text{£}168.3\text{k}$$

$$TV = 168.3 / (0.08 - 0.02) = 168.3 / 0.06 = \text{£}2,805\text{k}$$

Step 3 — Discount each cash flow correctly.

Year	Cash Flow (£000)	Discount Factor $(1.08)^n$	PV (£000)
1	120	1.0800	111.1
2	135	1.1664	115.8
3	150	1.2597	119.1
4	165 + 2,805 = 2,970	1.3605	2,182.3

Step 4 — Sum the present values.

$$NPV = 111.1 + 115.8 + 119.1 + 2,182.3 = \text{£}2,528.3\text{k}$$

Step 5 — Sanity check. The terminal value dominates (as expected for a going-concern business), and the NPV is below the undiscounted sum of cash flows—both signs that the arithmetic is directionally correct.

The flawed quarterly-rate approach had produced £2,641k—a 4.5% overstatement that would have materially affected an investment decision. Correcting the period-rate mismatch is the single most impactful fix available.

